FOR COMPLETE BEGINNERS

Where did 50 cents go?

Complete Modern Java, episode 7: expressions and operators in Java and TypeScript

> Operators work on values, and the types of those values decide the result.

A $10 bill split between 4 people printed 2. Java lost nothing: two ints divided keep the whole part.

One double in the division gives 2.5. Then operator order, and the left to right trap.

In this episode

- What an expression is
- + - * / and the remainder %
- Integer division
- Operator order, left to right

Coming next

- Numbers and precision (episode 8)
- Division by zero (episode 8)
- += and ++ (with loops)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each line printed

7 / 2 and 7 % 2	`3 and 1`
total / people (10 and 4, ints)	`2`
10.0 / people	`2.5`
2 + 3 * 4 and (2 + 3) * 4	`14 and 20`
10 / 4 * 1.0 and 10 * 1.0 / 4	`2.0 and 2.5`
TypeScript 7 / 2 and 10 / 4	`3.5 and 2.5`

Every row was run, not predicted.

02 Integer division and the remainder

Seven sweets, two people: three each, one left over. 7 / 2 is 3 (the whole part, never rounded); 7 % 2 is 1.

One double in the division gives decimal division. Brackets change the order, never the types.

Bill.java

```
int total = 10;
int people = 4;
IO.println(total / people);        // 2
IO.println(10.0 / people);         // 2.5
double totalMoney = 10.0;
IO.println(totalMoney / people);   // 2.5
```

> Printed 2, 2.5, 2.5.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

03 TypeScript

One number type, so the decimal stays. Math.floor rounds down to the whole number below; the result is still a number.

ops.ts

```
console.log(7 / 2);                // 3.5
console.log(10 / 4);               // 2.5
console.log(2 + 3 * 4);            // 14
console.log(Math.floor(10 / 4));   // 2
```

> npx tsc --noEmit was quiet; node printed 3.5, 2.5, 14, 2.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

Read an expression like Java does

1. Which operator goes first? * / % before + -.
2. Same priority? Left to right.
3. Two ints divided? The whole part only.
4. Want decimals? Put a double in the division, early.
5. Challenge: predict 9 / 2, 9 % 2, 9 / 2.0, 2 + 3 * 4, 10 / 4 * 1.0.

Everything else

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The code and the challenge files: complete-modern-java
https://github.com/code-with-sam-dev/complete-modern-java

If this was useful, a like and a subscribe help more than you would think. And tell me in the comments which part of your workflow you want taken apart next.

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